

Introduction to Mobile IP and Addressing

Module 36

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Hello friends I welcome you all to the video lecture series of cyber security and information security in this lecture we will study about mobile IP and its related terms first of all please see the contents which I shall cover first I will cover the brief introduction about mobile IP and addressing after covering brief introduction about mobile IP and addressing I shall take up agents in Mobile IP after that I will cover the three phases used in Mobile communication and last I shall take up inefficiency in Mobile IP so now let us start mobile communication has received a lot of attention in last decade the interest in Mobile communication on the internet means that the IP protocol originally designed for stationary devices must be enhanced to allow the

use of mobile computers computers that move from one network to another addressing the main problem that must be solved in providing mobile communication using the IP protocol is addressing stationary hosts the original Ip addressing was based on the assumption that a host is stationary attached to one specific Network a router uses an IP address to route an IP datagram an IP address has two parts a prefix and a suffix the prefix Associates a host to a network for example the IP address 10.3.4 24x 8 defines a host attached to the network 10.0.0.0 by 8 this implies that a host in the internet does not have an address that it can carry with itself from one place to another the address is valid only when the host is attached to the

network if the network changes the address is no longer valid routers use this Association to route a packet they use the prefix to deliver the packet to the network to which the host is attached this scheme works perfectly with stationary hosts the IP addresses are designed to work with stationary hosts because part of address defines the network to which the host is attached mobile host when a host moves from one network to another the IP addressing structure needs to be modified several Solutions have been proposed changing the address one simple solution is to let the mobile host change its address as it goes to the new network The Host can use DHCP to obtain a new address to associate it with the new network this approach has several drawbacks first the

configuration files would need to be changed second each time the computer moves from one network to another it must be rebooted third the DNS tables need to be revised so that every other host in the internet is aware of the change fourth if the host roams from from one network to another during a transmission the data exchange will be interrupted this is because the ports and IP addresses of the client and the server must remain constant for the duration of the connection two addresses the approach that is more feasible is the use of two addresses the host has its original address called the home address and a temporary address called the care of address as per figure the home address is permanent it Associates the host to its home network the network that is

permanent home of the host the care of address is temporary when a host moves from one network to another the care of address changes it is associated with the foreign Network the network to which the host moves mobile IP has two addresses for a mobile host one home address and one care off address the home address is permanent the care of address change as the mobile host moves from one network to another agents to make the change of address transparent to rest of the internet requires a home agent and a foreign agent figure shows the position of a home agent relative to the home network and a foreign agent relative to foreign Network we have shown the home and the foreign agent as routers but we need to emphasize that the

specific functions as an agent is performed in the application layer in other words they are both routers and host home agent the home agent is usually a router attached to the home network of the mobile host the home agent acts on behalf of the mobile host when a remote host sends a packet to the mobile host the home agent receives the packet and sends it to the foreign agent foreign agent the foreign agent is usually a router attached to the foreign Network the foreign agent receives and delivers package sent by the Home agent to the mobile host the mobile host can also act as a foreign agent in other words the mobile host and foreign agent can be the same however to do this a mobile host must be able to receive a care of address by itself which can be done through using the

DHCP in addition the mobile host needs the necessary software to allow it to communicate with the home agent and to have two addresses its home address and its care of address this dual addressing must be transparent to the application programs when the mobile host acts as a foreign agent the care of address is called a colocated care of address when the mobile host and the foreign agent are the same the care of address is called a collocated care of address the advantage of using a c ated care of addresses that the mobile host can move to any network without worrying about the availability of a foreign agent three phases in Mobile communication to communicate with a remote host a mobile host goes through three phases agent Discovery registration and data transfer as shown

in the figure the first phase agent Discovery involves the mobile host the foreign agent and the home agent the second phase registration also involves the mobile host and the two agents finally in the third phase the remote host is also involved we discuss each phase separately agent Discovery the first phase in Mobile communication agent Discovery consists of two surfaces a mobile host must discover a home agent before it leaves its home network a mobile host must also discover a foreign agent after it has moved to a foreign Network this discovery consists of learning the care of address as well as the foreign agent address the discovery involves two types of messages advertisement and solicitation agent advertisement when a router advertises its presence on a network using an icmp

router advertisement it can append an agent advertisement to the packet if it acts as an agent figure shows how an agent advertisement is piggy bagged to the router advertisement packet the field descriptions shown in figure are as follows type the 8bit type field is set to 16 length the 8bit length field defines the total length of the extension message not the length of the icmp advertisement message sequence number the 16bit sequence number field holds the message number the recipient can use the sequence number to determine if a message is lost lifetime the lifetime field defines the number of seconds that the agent will accept requests if the value is a string of ones the lifetime is in finite code the

code field is an 8bit flag in which each bit is set one or unset to zero care of addresses this field contains a list of addresses available for use as care of addresses the mobile host can choose one of these addresses the selection of this SC of address is announced in the registration request note that this field is used only by a foreign agent agent solicitation when a mobile host has moved to a new network and has not received agent advertisements it can initiate an agent solicitation it can use the icmp solicitation message to inform an agent that it needs assistance registration the second phase in Mobile communication is registration after a mobile host has moved to a foreign Network and discovered the foreign agent it must register there are four aspects

of registration first the mobile host must register itself with the foreign agent second the mobile host must register itself with its home agent this is normally done by the foreign agent on behalf of the mobile host third the mobile host must renew registration if it has expired fourth the mobile host must cancel its registration when it returns home request and reply to register with the foreign agent and the home agent the mobile host uses a registration request and a registration reply registration request a registration request is sent from the mobile host to the foreign agent to register its care of address and also to announce its home address and home agent address the foreign agent after receiving and registering the request relays the message to the home agent note that the

home agent now knows the address of foreign agent because the IP packet that is used for relaying has the IP address of the foreign agent as the source address figure shows the format of the registration request the field descriptions shown in figure are as follows type the the

8bit type field defines the type of message for a request message the value of this field is one flag the 8bit flag field defines forwarding information the value of each bit can be set or unset lifetime this field defines the number of seconds the registration is valid if the field is a string of zeros the request message is asking for D registration if the field is a string of once the lifetime is infinite home address this field contains the permanent address of the mobile host mobile agent address this

field contains the address of the home agent care of address this field is the temporary address of mobile host identification this field contains the 64bit number that is inserted into the request by the mobile host and repeat in the reply message it matches a request with a reply extensions variable length extensions are used for authentication they allow a home agent to authenticate the mobile agent registration reply a registration reply is sent from the home agent to the foreign agent and then relay to the mobile host the reply confirms or denies the registration request data transfer after agent Discovery and registration a mobile host can communicate with the remote host figure shows the process of data

transfer I'll explain one by one from remote host to home agent path one of figure shows this step when a remote host wants to send a packet to the mobile host it uses its address as the source address and the home address of the mobile host as the destination address in other words the remote host sends a packet as though the mobile host is at its home network the packet however is intercepted by the Home agent which pretends it is the mobile Host this is done using the proxy ARP technique from home agent to foreign agent path two of the figures shows the step after receiving the packet the home agent sends the packet to the foreign agent using the tunneling concept the home agent encapsulates the whole IP packet inside another IP packet using its address as the source and the

foreign agent's address as the destination from foreign agent to mobile host part three of figure shows this step when the foreign agent receives the packet it removes the original packet however since the destination address is the home address of mobile host the foreign agent consults a registry table to find the care of address of the mobile host otherwise the packet would just be sent back to the home network the packet is then sent to the care of address from mobile host to remote host path four of figure shows the step when a mobile host wants to send a packet to a remote host for example a response to the packet it has received it sends as it does normally the mobile host prepares a packet with its home address as the source and the address of remote host as the destination although the packet comes

from the foreign Network it has the home address of the mobile host transparency in this data transfer process the remote host is unaware of any Movement by the mobile host the remote host send packets using the home address of the mobile host as the destination address it receives the packets that have the home address of the mobile host as the source address the movement is totally transparent the rest of the internet is not aware of the mobility of mobile host inefficiency in Mobile IP communication involving mobile IP can be inefficient the

inefficiency can be severe or moderate this severe case is called double crossing or 2X the moderate case is called triangle routing or dog leg routing double crossing according to

figure double crossing occurs when a remote host communicates with the mobile host that has moved to the same network as the remote host when the mobile host sends a packet to the remote host there is no inefficiency the communication is local however when the remote host sends a packet to the mobile host the packet crosses the internet twice since a computer usually communicates with other local computers the inefficiency from double crossing is significant triangle routing triangle routing occurs when the remote host communicates with the mobile host that is not attached to the same network or site as the mobile host when the mobile host sends a packet to the remote host there is no inefficiency

however when the remote host sends a packet to the mobile host the packet goes from remote host to the home agent and then to the mobile host the packet travels the two sides of a triangle instead of just one side one solution to inefficiency is for the remote host to bind the care of address to the home address of a mobile host for example when a home agent receives the first packet for a mobile host it forwards the packet to the foreign agent it could 18 minutes, 1 second also send an update binding packet to the remote host so that future packets to this host can be sent to the care of address the remote host can keep this information in a cache the problem with this strategy is that the cash entry

becomes outdated once the mobile host moves In this case the home agent needs to send a warning packet to the remote host to inform it of the change so friends this was all about mobile IP and its related terms hope the concepts explained in this lecture were understandable and helpful hope to see you in next lecture till then goodbye thank you